

[SP-11]

2. LLN & CLT

1. WIS: CLT implies WLLN.

According to CLT,

$$\frac{\sqrt{n}}{\sigma}(\bar{X}_n - \mu) \rightarrow N(0, 1) \text{ in distribution} \quad \checkmark$$

Since $\sqrt{n} \rightarrow \infty$, $\bar{X}_n - \mu$ shall approach zero to prevent the whole term from exploding. ✓

CLT talks about convergence in dist.
WLLN states that $\bar{X}_n$ is extremely close to μ with extremely high Prob.

2.

(a). WIS: $\text{Pois}(n)$ approximated by Normal.

Let $X \sim \text{Pois}(n)$ where n is a positive integer.

Then, we can express X as ✓

$X = X_1 + \dots + X_n$,
where $X_i \stackrel{\text{i.i.d.}}{\sim} \text{Pois}(1)$. $\Rightarrow E[X_i] = \text{Var}(X_i) = 1$

$$P(a \leq X \leq b) = P\left(\frac{a-n}{\sqrt{n}} \leq \frac{X-n}{\sqrt{n}} \leq \frac{b-n}{\sqrt{n}}\right).$$

We observe that

$$\frac{X-n}{\sqrt{n}} = \frac{\sum_{i=1}^n X_i - n \cdot 1}{\sqrt{n} \cdot 1} \sim N(0, 1)$$

in distribution when n is large.

Hence,

$$P(a \leq X \leq b) \approx \Phi\left(\frac{b-n}{\sqrt{n}}\right) - \Phi\left(\frac{a-n}{\sqrt{n}}\right).$$

And thus

$$\sum_{j=1}^n X_j = X \rightarrow N(n, n)$$

in distribution. ✓

(b). Given $N \sim \text{Pois}(n)$ where n is a positive integer.

$$P(N=n) = \frac{n^n e^{-n}}{n!}.$$

Based on (a),

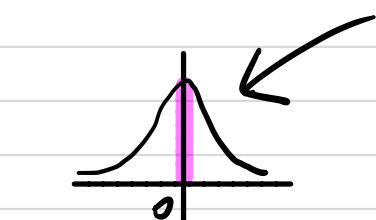
$$P(N=n) = P\left(n - \frac{1}{2} < N < n + \frac{1}{2}\right)$$

$$\approx \Phi\left(-\frac{1}{2\sqrt{n}}\right) - \Phi\left(\frac{1}{2\sqrt{n}}\right)$$

$$= \int_{-\frac{1}{2\sqrt{n}}}^{\frac{1}{2\sqrt{n}}} \frac{1}{\sqrt{2\pi}} e^{-\frac{t^2}{2}} dt$$

$$\approx \frac{1}{\sqrt{2\pi}} \int_{-\frac{1}{2\sqrt{n}}}^{\frac{1}{2\sqrt{n}}} 1 dt$$

$$= \frac{1}{\sqrt{2\pi n}}.$$



$\left. \begin{array}{l} \text{\textit{n} is large,} \\ \frac{1}{2\sqrt{n}} \rightarrow 0, \\ e^{-\frac{t^2}{2}} \rightarrow e^0 \rightarrow 1 \end{array} \right\}$

$$\left. \begin{array}{l} \frac{1}{2\sqrt{n}} \rightarrow 0, \\ e^{-\frac{t^2}{2}} \rightarrow e^0 \rightarrow 1 \end{array} \right\}$$

Hence,

$$\frac{n^n e^{-n}}{n!} \approx \frac{1}{\sqrt{2\pi n}}$$

$$\Rightarrow n! \approx \sqrt{2\pi n} \left(\frac{n}{e}\right)^n$$

$\Rightarrow$ Stirling's formula. ✓

3.

Because the degrees of freedom $m \geq 3$, both expectation and variance exist.

Let $T_j = z_j / \sqrt{V_j/m}$ where

$z_j \sim N(0,1)$ and $V_j \sim \chi^2(m)$ (Gamma($\frac{m}{2}, \frac{1}{2}$))

z_j and V_j are indep.

$$E[T_j] = E[z_j] \cdot E[\sqrt{V_j/m}] = 0$$

$$E[T_j^2] = E[z_j^2 / (V_j/m)]$$

$$= E[z_j^2] \cdot m E[1/V_j]$$

$$E[z_j^2] = \text{Var}(z_j) + (E[z_j])^2 = 1.$$

$$E[1/V_j] = \int_0^\infty \frac{1}{x} \frac{1}{\Gamma(\frac{m}{2})} (\frac{1}{2}x)^{\frac{m}{2}-1} e^{-\frac{1}{2}x} \frac{1}{x} dx$$

$$= \frac{1}{2} \frac{\Gamma(\frac{m}{2}-1)}{\Gamma(\frac{m}{2})} \int_0^\infty \frac{1}{\Gamma(\frac{m}{2}-1)} (\frac{1}{2}x)^{\frac{m}{2}-2} e^{-\frac{1}{2}x} \frac{1}{x} dx$$

$$= \frac{1}{2} \frac{\Gamma(\frac{m}{2}-1)}{\Gamma(\frac{m}{2})} = \frac{1}{m-2}$$

$$\Rightarrow E[T_j^2] = \frac{m}{m-2}$$

$$\Rightarrow \text{Var}(T_j) = E[T_j^2] - (E[T_j])^2 = \frac{m}{m-2} \quad \checkmark$$

To find a_n, b_n s.t. $a_n(T_1 + \dots + T_n - b_n)$

$\rightarrow N(0,1)$ in distribution.

$$\begin{cases} E[a_n(T_1 + \dots + T_n - b_n)] = 0 \\ \text{Var}(a_n(T_1 + \dots + T_n - b_n)) = 1 \end{cases}$$

And we get

$$\begin{cases} a_n = \sqrt{\frac{n-2}{nn}} \\ b_n = 0 \end{cases}$$



4.

$$\frac{\sum_{j=1}^n X_j}{\sum_{j=1}^n Y_j} = \frac{\frac{1}{n} \sum_{j=1}^n X_j}{\frac{1}{n} \sum_{j=1}^n Y_j} = \frac{\bar{X}_n}{\bar{Y}_n}$$

By SLLN, as $n \rightarrow \infty$, $\bar{X}_n \rightarrow 2$, $\bar{Y}_n \rightarrow 3$ with Prob. 1

$$\Rightarrow \frac{X_1 + X_2 + \dots + X_n}{Y_1 + Y_2 + \dots + Y_n} \rightarrow \frac{2}{3} \text{ with Prob. 1.}$$

↳ Note that if two events A, B both have Prob. 1, then $A \cap B$ has Prob. 1. }

We don't need to assume that X_i are independent of Y_j .



5. Let I_j be the indicator r.v. such that

$$I_j = \begin{cases} 1 & \text{if } (X_j, Y_j) \text{ is in the area below } f(x). \\ 0 & \text{otherwise.} \end{cases}$$

$$P(I_j = 1) = \frac{1}{c(b-a)} \int_a^b f(x) dx.$$

Since I_j 's are i.i.d., by LLN,

$$\frac{1}{n} \sum_{j=1}^n I_j \rightarrow E[I_j] = \frac{1}{c(b-a)} \int_a^b f(x) dx \text{ as } n \rightarrow \infty.$$

Therefore,

$$\int_a^b f(x) dx \approx c(b-a) \cdot \left(\frac{1}{n} \sum_{i=1}^n z_i \right)$$

$$= (\text{Area of rectangle}) \times$$
$$(\text{ratio of points in area below } y=f(x))$$



II. Multivariate Normal

1.

Pf:

Given (X_1, X_2, X_3) is MVN,

$$t_1 X_1 + t_2 X_2 + t_3 X_3$$

is Normal for any t_1, t_2, t_3 .

Let $t_3 = 0$. For any t_1, t_2 ,

$$t_1 X_1 + t_2 X_2$$

is Normal.

$\Rightarrow (X_1, X_2)$ is MVN.



2. True.

Pf:

Given $(X_1, \dots, X_n)$ and $(Y_1, \dots, Y_m)$ are MVN, for any $(t_1, \dots, t_n)$ and $(s_1, \dots, s_m)$,

$$t_1 X_1 + \dots + t_n X_n \text{ is Normal}$$

$$s_1 Y_1 + \dots + s_m Y_m \text{ is Normal}$$

Since X is indep. of Y ,

$(t_1 X_1 + \dots + t_n X_n) + (s_1 Y_1 + \dots + s_m Y_m)$ is Normal.

$\Rightarrow (X_1, \dots, X_n, Y_1, \dots, Y_m)$ is MVN.



3.

WTF: a, b, c, d s.t. $Z = aX + bY$ and $W = cX + dY$ are indep. $N(0,1)$

Given the problem statement,

$$\left\{ \begin{array}{l} \text{Cov}(Z, W) = 0 \\ E[Z] = 0 \\ E[W] = 0 \\ \text{Var}(Z) = 1 \\ \text{Var}(W) = 1 \end{array} \right.$$

$$\Rightarrow \left\{ \begin{array}{l} ac + bd + (ad + bc)\rho = 0 \\ a^2 + 2ab\rho + b^2 = 1 \\ c^2 + 2cd\rho + d^2 = 1 \end{array} \right.$$

Four variables, three equations.

Let $a=0$. Then, $b=1$, $d=-c\rho$.

$$c = 1/\sqrt{1-\rho^2}$$

$$\Rightarrow \left\{ \begin{array}{l} a=0 \\ b=1 \\ c=1/\sqrt{1-\rho^2} \\ d=-\rho/\sqrt{1-\rho^2} \end{array} \right.$$



4. $\bar{X}_n = \frac{1}{n} \sum_{j=1}^n X_j$

$$S_n^2 = \frac{1}{n-1} \sum_{j=1}^n (X_j - \bar{X}_n)^2.$$

WTS: $\bar{X}_n$ and S_n^2 are independent.

Pf.: Considering random vector
 $(\bar{X}_n, X_1 - \bar{X}_n, \dots, X_n - \bar{X}_n)$, any linear combination

$t_0 \bar{X}_n + t_1(X_1 - \bar{X}_n) + \dots + t_n(X_n - \bar{X}_n)$
is a linear combination of $X_1, \dots, X_n$.

Since $X_1, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$,

$\Rightarrow t_0 \bar{X}_n + t_1(X_1 - \bar{X}_n) + \dots + t_n(X_n - \bar{X}_n)$
is Normal for any $t_0, t_1, \dots, t_n$.

$\Rightarrow (\bar{X}_n, X_1 - \bar{X}_n, \dots, X_n - \bar{X}_n)$ is MVN.

And,

$$\begin{aligned} & \text{Cov}(\bar{X}_n, X_j - \bar{X}_n) \\ &= \text{Cov}(\bar{X}_n, X_j) - \text{Var}(\bar{X}_n) \\ &= \text{Cov}\left(\frac{1}{n} X_j, X_j\right) - \text{Var}(\bar{X}_n) \\ &= \frac{1}{n} \sigma^2 - \frac{1}{n} \sigma^2 = 0. \end{aligned}$$

$\Rightarrow \bar{X}_n$ and $X_j - \bar{X}_n$ are uncorrelated

$\Rightarrow \bar{X}_n$ and $X_j - \bar{X}_n$ are indep.

$\Rightarrow \bar{X}_n$ and function of $X_j - \bar{X}_n$ are indep.

Since

S_n^2 is a fcn of $(X_1 - \bar{X}_n, \dots, X_n - \bar{X}_n)$,

$\Rightarrow \bar{X}_n$ and S_n^2 are indep. 

III. Markov Chains

1. Given $X_0, X_1, X_2 \dots$ be a Markov chain,
let Q be the Transition matrix.
 $\Rightarrow X_0, X_2, X_4 \dots$ is a Markov chain
with transition matrix Q^2 .



This makes sense. To predict X_{10}
given $X_0, X_2, \dots, X_8$, we only use
 X_8 , the most recent one.

2.

(a).

$$Q = \begin{pmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.25 & 0.75 & 0 & 0 \\ 0 & 0 & 0.25 & 0.75 \\ 0 & 0 & 0.75 & 0.25 \end{pmatrix}$$



- (b). Recurrent states: 1, 2, 3, 4.
no transient state.



- (c). Solving $SQ = S$,

we get

$$S = \left(\frac{1}{3}, \frac{2}{3}, 0, 0\right) \text{ or}$$

$$S = \left(0, 0, \frac{1}{2}, \frac{1}{2}\right).$$

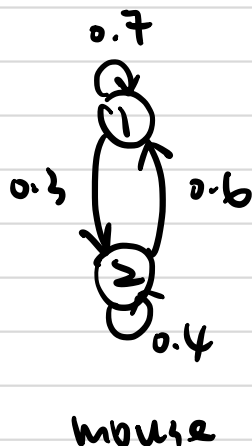
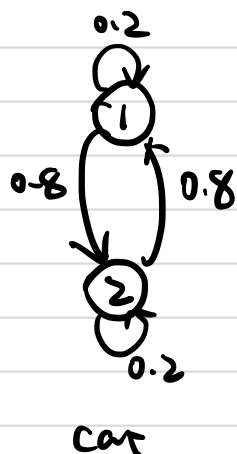


And

$$+ t\left(\frac{1}{3}, \frac{2}{3}, 0, 0\right) + (1-t)\left(0, 0, \frac{1}{2}, \frac{1}{2}\right)$$

for any $0 \leq t \leq 1$.

3.
(a).



For cat,

$$Q_c = \begin{pmatrix} 0.2 & 0.8 \\ 0.8 & 0.2 \end{pmatrix}$$

$$S_c = \begin{pmatrix} 0.5 & 0.5 \end{pmatrix} \text{ by symmetry } \checkmark$$

For mouse

$$Q_m = \begin{pmatrix} 0.7 & 0.3 \\ 0.6 & 0.4 \end{pmatrix}$$

$$S_m = \begin{pmatrix} \frac{2}{3} & \frac{1}{3} \end{pmatrix} \checkmark$$

(b). Yes, $Z_0, Z_1, Z_2, \dots$ is a Markov chain.
The next (cat, mouse) position is indep.
of past (cat, mouse) positions given
the current (cat, mouse) position. $\checkmark$

4.

With the problem statement, we can construct "random walk on undirected graph". King being on a position is a state of this Markov chain.

We can classify positions into three types: center ones (8 degs); edge ones (5 degs) and corner ones (3 degs)

⇒ The stationary distribution of a chain is $8a, 5a, 3a$

$$\Rightarrow 8a \cdot 36 + 5a \cdot 24 + 3a \cdot 4 = 1$$

$$\Rightarrow a = 1/420$$

$$\Rightarrow 2/105, 1/84, 1/140$$

5.

(a). The prob. of going from a permutation g to a permutation h is $1/\binom{26}{2}$ if h can be obtained from g in one swap; otherwise 0.

The chain can be represented by "random walk on undirected graph" with each node has degrees $\binom{26}{2}$.

⇒ each entry of stationary distribution is $\binom{26}{2} / [26! \cdot \binom{26}{2}] = 1/26!$

(b).

Let $q(g, h)$ be the prob. of going from g to h for any $g \neq h$.

We let $p(g, h)$ be the prob. of proposing h given g . When $s(g) \leq s(h)$,
 $q(g, h) = p(g, h)$

And,
 $q(h, g) = p(h, g) \cdot \frac{s(g)}{s(h)}$

Since $p(h, g) = p(g, h) = q(g, h)$

$$\Rightarrow q(h, g) = q(g, h) \frac{s(g)}{s(h)}$$

$$\Rightarrow s(g) q(g, h) = s(h) q(h, g)$$

When $s(g) > s(h)$.

$$q(g, h) = p(g, h) \frac{s(h)}{s(g)}$$

And,
 $q(h, g) = p(h, g) = p(g, h)$

$$\Rightarrow s(g) q(g, h) = s(h) q(h, g)$$

$\Rightarrow$ The chain is reversible.

And, stationary distribution is proportional to the list of all scores $s(g)$.



} An example of Metropolis's

Algorithm $\Rightarrow$ A powerful, general method of transmuting one

Markov chain into a new one with a certain desired stationary distribution. {